

Unit 9, Lesson 5: Equivalent Numerical Expressions

Benchmarks

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

Learning Target

- ☐ I can determine if numerical expressions are equivalent.

9.5.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong? Justify your reasoning.

$(13 - 12 \div 2) + 3 \times 2$	$12 + (9 \div 3)^2$
$20 - 16 \div 8 + 5$	$38 - (2 + 4) + 20 - 3$

9.5.2 Exploration: Equivalent Numerical Expressions

1. Is $14 - \frac{3}{4}(12)$ equivalent to $\frac{47}{5} - \left(\frac{44}{5} \div 2\right)$? Explain how you determined your answer.
2. Consider the expression $-\frac{3}{4}(12)$.
 - a. What is the value of $-\frac{3}{4}(12)$?
 - b. Explain if $-\frac{3}{4}(12)$ can be rewritten as $0.75(-12)$?
 - c. Does $\frac{-3(12)}{4}$ have the same value as the previous expressions?
 - d. Create another expression equivalent to $-\frac{3}{4}(12)$. Share your answer with your partner.

The expressions in the previous problem all have the same value, but they are written differently.

3. What do you notice as the similarities and differences between $-\frac{3}{4}(12)$, $0.75(-12)$, and $\frac{-3(12)}{4}$?
4. Explain which representation is easiest for you to evaluate using mental math.

9.5.3 Guided Instruction: Equivalent Numerical Expressions



Numerical expressions are equivalent if their values, when simplified, are the same.

Algebraic expressions are equivalent if they have the same value no matter what number is substituted for each of the variables.

1. Are the following expressions equivalent? Show your work or explain your thinking.

$$\frac{9}{4} + \left(3.5 - 7\frac{1}{2}\right) \text{ and } \left(-\frac{1}{2} \cdot \frac{3}{4} + 6.5\right) - 7\frac{7}{8}$$

2. Work with your partner to create a third expression that is also equivalent to $\frac{9}{4} + \left(3.5 - 7\frac{1}{2}\right)$.

9.5.4 Try It: Equivalent Numerical Expressions

1. Evaluate expressions A-L.

A $-\frac{1}{3}(-24) + 3$	B $(-2.25 \times 4) \cdot 2$	C $-\frac{15}{15} + 4\frac{1}{3}$
D $3(2.5 - 6) + 7\frac{1}{2}$	E $4(-\frac{1}{4})(1)(-1^2)$	F $3 + \frac{-24}{-3}$
G $\frac{5(-4\frac{1}{4} + 6.25)}{3}$	H $-6\frac{2}{3} \cdot 2.4 \div (-4)$	I $-7.5 + (6\frac{1}{6} \cdot 3)$

J $\left(-4.2 + 9\frac{4}{5}\right) + \left(1 + 4\frac{4}{10}\right)$	K $-\left(\frac{-18}{-5}\right) - 0.4$	L $\left -3\frac{1}{8} - 0.875\right $
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2. Sort the expressions into groups based on equivalence. Record the letters of the equivalent expressions.

Equivalent Expressions

9.5.5 Collaboration: Generating Equivalent Expressions

1. Work with a partner to choose a numerical value.

Individually, create a unique expression using rational numbers that equals the numerical value chosen when evaluated. Include a **minimum** of each of the following:

- four operations
- one grouping symbol
- two rational numbers written as fractions

When you are finished creating your expression, trade and check each other’s numeric expressions.

Numerical Value _____	
Your Expression	Your Partner’s Expression

2. Are they equivalent? If not, correct any errors.

9.5.6 Wrap-Up: Cool Down

1. Are the following expressions equivalent? Show your work or explain your reasoning.

$$\left(-7.6 + 2\frac{1}{2}\right)(-6) \text{ and } (-8)\left(-3\frac{1}{10}\right) + \frac{2}{3}(8.6)$$